

- (c) Twee bronne straal golwe uit met golflengte λ . Die een bron is op afstand L vanaf 'n punt P en die ander een se afstand is $L + \Delta L$.
Two sources emit waves with wavelength λ . The one source is a distance L from point P and the other one is $L + \Delta L$ from P .

- (i) Gee die fase ϕ in terme van 'n getal n vir interferensie by P vir:
Give the phase ϕ as function of a number n for interference at P for:
 Konstruktiewe interferensie / Constructive interference:

$$\phi = n (2\pi) \quad \checkmark$$

Destruktiewe interferensie / Destructive interference

$$\phi = \left(n + \frac{1}{2}\right) 2\pi \quad \checkmark$$

- (ii) Skryf nou ΔL in terme van n vir: / Write ΔL in terms of n for:

Konstruktiewe interferensie / Constructive interference

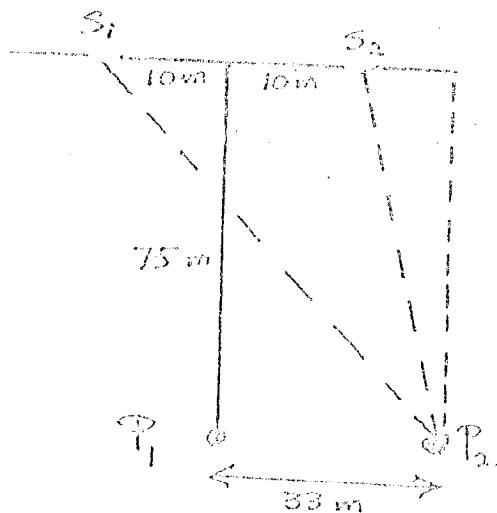
$$\Delta L = n \lambda \quad \checkmark$$

Destruktiewe interferensie / Destructive interference

$$\Delta L = \left(n + \frac{1}{2}\right) \lambda \quad \checkmark$$

- (d)(i) 'n Brekwater muur in die see het twee openinge S_1 en S_2 wat 20 m van mekaar is. Bepaal uit gegewens in die skets of 'n persoon in 'n boot by punt P_1 konstruktiewe of destruktiewe interferensie ondervind. (Die faseverskil by P is $\phi = \frac{\Delta L}{\lambda} 2\pi$).
(2)

Ocean waves pass through two openings S_1 and S_2 in a breakwater. The openings are 20 m apart. Determine from the values in the figure if a person in a boat at P_1 experience constructive or destructive interference. (The phase difference of P is $\phi = \frac{\Delta L}{\lambda} 2\pi$).



$$\phi = \frac{\Delta L}{\lambda} 2\pi = \frac{0}{\lambda} 2\pi \quad \checkmark \checkmark$$

$$\phi = 0 \quad \checkmark$$

at P_1 there will be constructive interference. (2)

(4)

Vraag / Question 5(d)(ii)(Vervolg / Continue) /